Project name: ***Differential Amplifier with resistive loads using transistors***

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Course Vendor: ***L&T Edutech***

Design of Schematic and Simulation of Differential Amplifier Using Electric EDA Tool

Aim of the Project

The objective of this project is to design and simulate a CMOS differential amplifier with resistive loads using MOSFET transistors. Through this project we will gain hands-on experience in analog circuit design, understand differential signaling principles, and analyze key performance parameters such as differential gain and common-mode rejection ratio (CMRR). The project also involves schematic creation, SPICE-based simulation, layout design, and post-layout verification using the Electric VLSI EDA tool, thereby bridging the gap between theoretical knowledge and practical VLSI design flow.

Project Design Specification

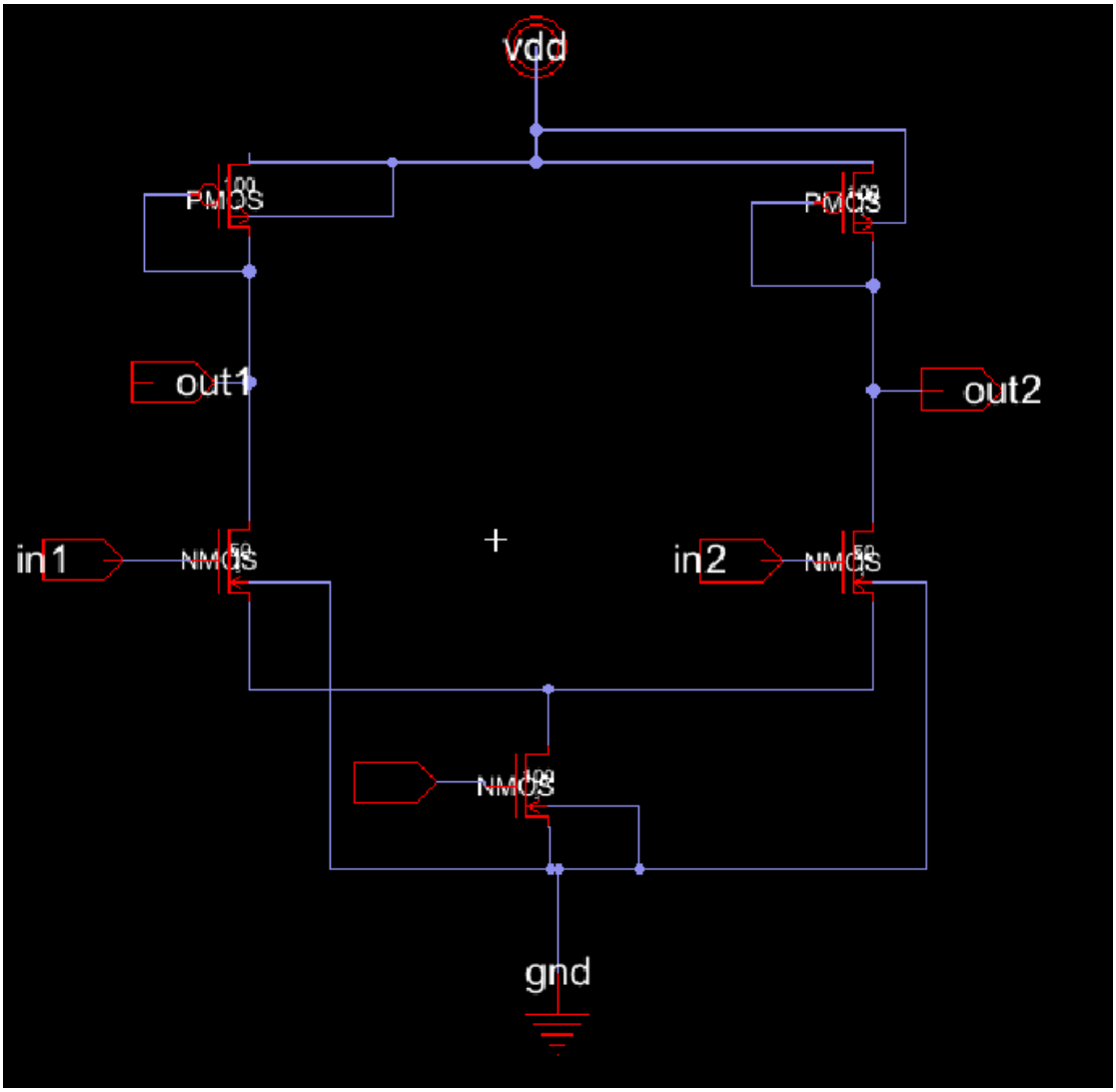
- **Technology**: 300nm CMOS process (MOSIS process)
- **Power Supply**: VDD = 5V, VSS = 0V
- **Input Type**: Differential inputs (VIN1 and VIN2)
- **Load Type**: Resistive load at the drain of PMOS transistors
- **Tail Current Source**: Implemented using NMOS biased by a DC voltage
- **Expected Output**: Differential voltage swing at the outputs (Vout1 and Vout2)
- **Tools Used**: Electric VLSI for schematic and layout design, LTspice for simulation

Schematic Diagram

The differential amplifier circuit consists of:

- **Two PMOS transistors (M1 and M2)** as resistive loads connected to VDD
- **Two NMOS transistors (M3 and M4)** forming the differential pair
- **A tail NMOS transistor (M5)** controlled by a bias voltage VTAIL
- **Inputs VIN1 and VIN2** driven by pulse sources
- **Output nodes Vout1 and Vout2** taken from the drain of M1 and M2 respectively

The schematic was created using Electric VLSI tool by placing NMOS and PMOS transistors, adding supply voltages, and connecting them appropriately to form a balanced differential pair.



Spice Code:

```
* Power Supplies
VDD vdd 0 DC 5
VSS 0 0 DC 0
* Differential Inputs
VIN1 in1 0 PULSE(0 0.2 0n 1n 1n 10n 20n)
VIN2 in2 0 PULSE(0.2 0 0n 1n 1n 10n 20n)
* Tail Bias Input
VTAIL in3 0 DC 0.9
* PMOS Active Loads
M1 out1 in1 vdd vdd PMOS
M2 out2 in2 vdd vdd PMOS
* NMOS Differential Pair
M3 out1 in1 n1 0 NMOS
M4 out2 in2 n1 0 NMOS
* NMOS Tail Transistor
M5 n1 in3 0 0 NMOS
* Include C5 Model File
.include C:\Users\Jagruthi K\Downloads\C5_models.txt
* Simulation Command
.TRAN 0.1n 100n
* Probes
.PRINT TRAN V(out1) V(out2)
*.PROBE
.END
```

Schematic Simulation Results

The simulation was performed in LTspice using the exported SPICE netlist from Electric.

Test Cases:

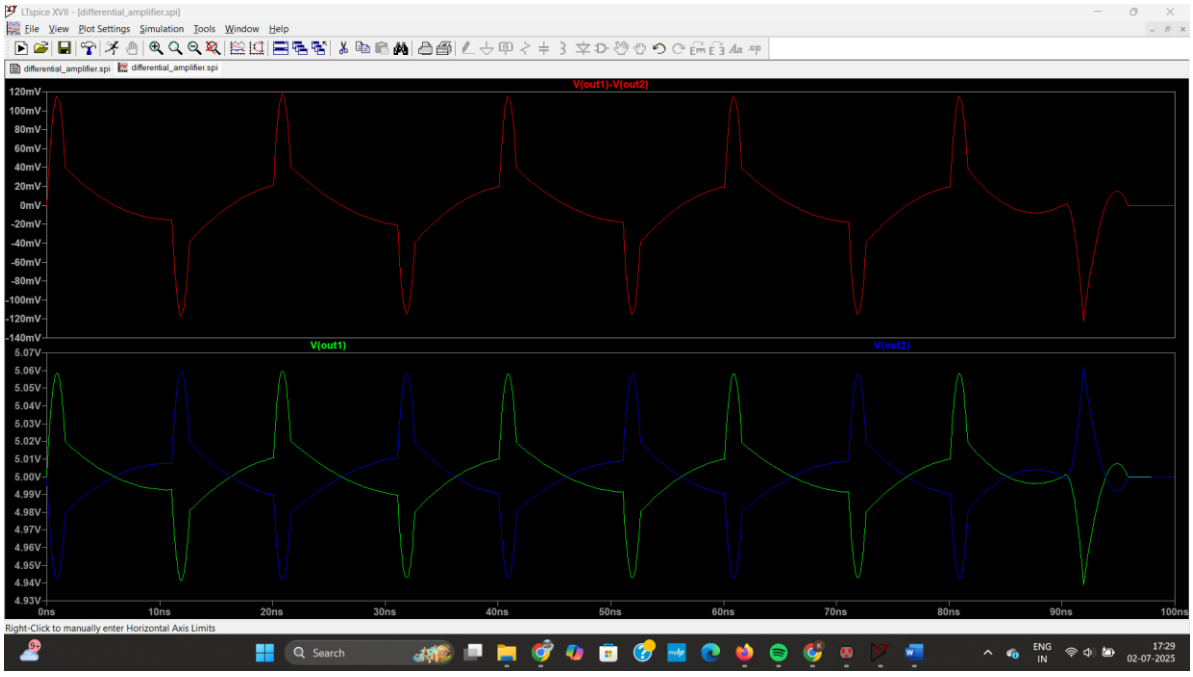
- Differential mode: VIN1 = PULSE(0 0.2V), VIN2 = PULSE(0.2 0V)
- Common-mode test: VIN1 = VIN2 = PULSE(0 0.2V)

Results:

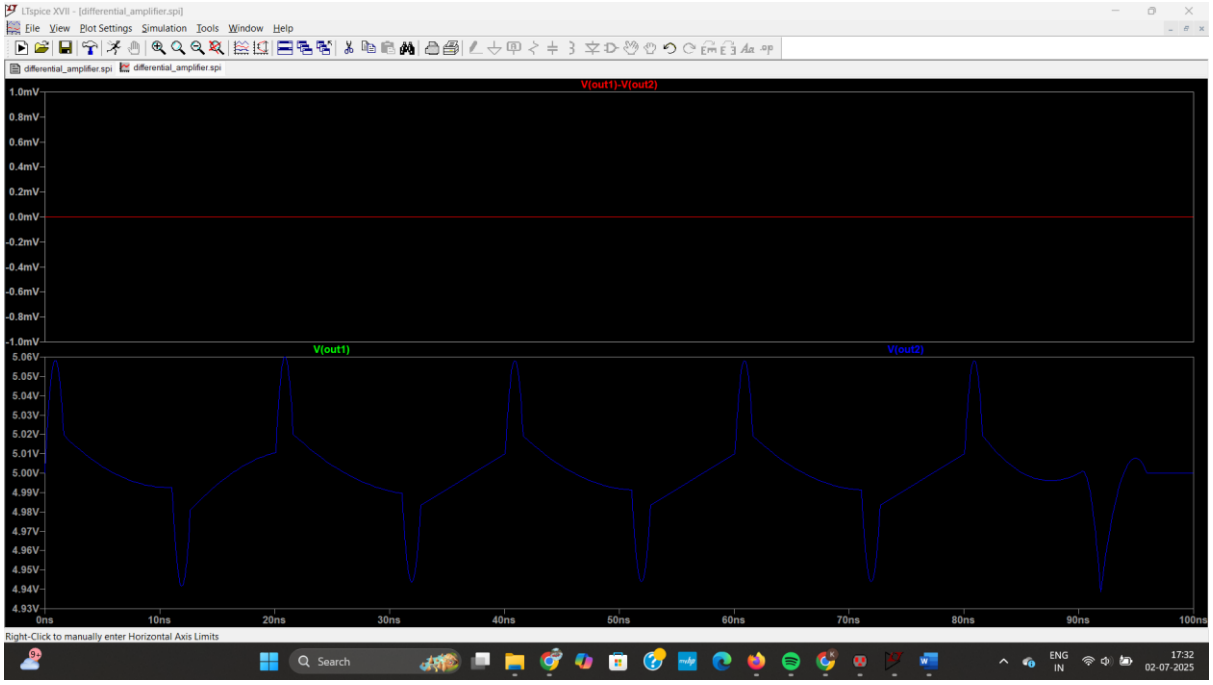
- In differential mode, a clear voltage difference is observed between Vout1 and Vout2.
- In common-mode, both outputs remain nearly constant indicating good common-mode rejection.

Waveforms: Time vs. V(out1), V(out2), and differential voltage (Vout1 - Vout2)

Differential Mode:



Common Mode:



<u>Differential Gain (Ad)</u>

(From the first image — differential mode test)

- From the red waveform V(out1) - V(out2) (top plot):
  Output differential swing = ±100 mV → → So, ΔVout = 200 mV (peak-to-peak)
- From the input definition (PULSE sources, as mentioned earlier):
  Input differential= VIN1 - VIN2 = 0.2 V (peak-to-peak)

➤ Differential Gain: Ad = ΔVout/ ΔVin = 0.2/0.2 = 1

<u>Common-Mode Gain (Ac)</u>

(From the second image — common mode test)

- VIN1 = VIN2 = Pulse(0 to 0.2V) → input swing = 0.2V peak-to-peak (common-mode)
- From the red waveform V(out1) - V(out2) in the top plot:
  It remains flat near 0V, i.e., only ±2mV max

➤ Common-Mode Gain: Ac = ((Vout1+Vout2)/2)/Vin = 0.004/0.2 = 0.02

<u>CMRR Calculation</u>

➤ Linear: Ad/Ac = 1/0.02 = 50

➤ In decibels (dB): 20 log(50) = 33.98 dB = 34 dB

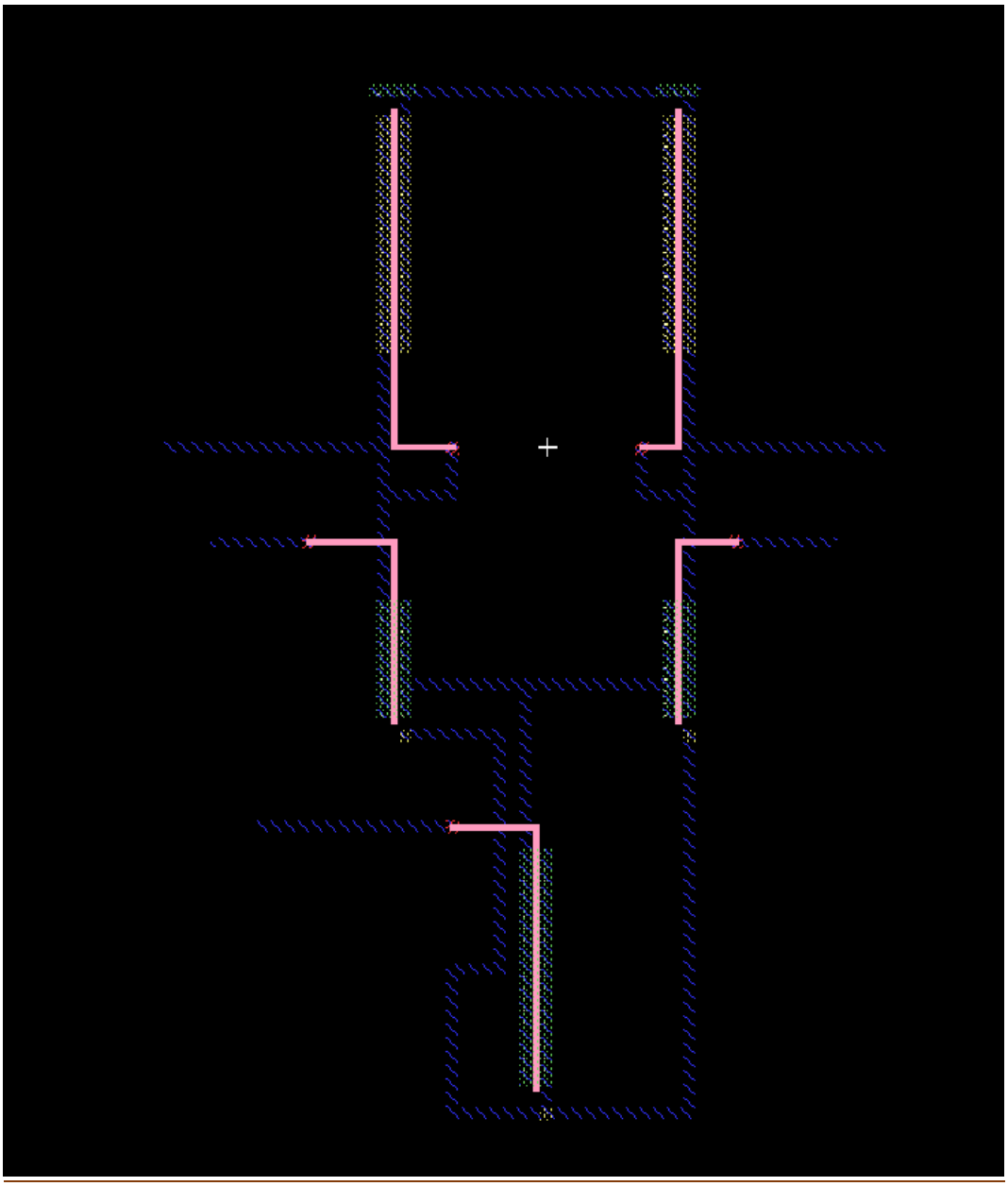
Design of Layout and Simulation of Differential Amplifier Using Electric EDA Tool

Layout

The layout was created using Electric EDA by:

- Placing PMOS and NMOS transistors from the component library
- Routing metal and poly connections using standard CMOS layout practices
- Ensuring minimal area and parasitic effects
- Following DRC rules of MOSIS technology

Layout Diagram:



Layout Simulation Results

- **Post-layout extraction** was done using Electric to generate the netlist.
- The extracted netlist was simulated in LTspice.
- Results were compared with schematic simulation and matched closely.
- Slight variations were due to parasitics introduced in layout.

Spice code:

```
* Power Supplies

VDD vdd 0 DC 5

VSS 0 0 DC 0

* Differential Inputs

VIN1 in1 0 PULSE(0 0.2 0n 1n 1n 10n 20n)

VIN2 in2 0 PULSE(0.2 0 0n 1n 1n 10n 20n)

* Tail Bias Input

VTAIL in3 0 DC 0.9

* PMOS Active Loads

M1 out1 in1 vdd vdd PMOS

M2 out2 in2 vdd vdd PMOS

* NMOS Differential Pair

M3 out1 in1 n1 0 NMOS

M4 out2 in2 n1 0 NMOS

* NMOS Tail Transistor

M5 n1 in3 0 0 NMOS

* Include C5 Model File

.include C:\Users\Jagruthi K\Downloads\C5_models.txt

* Simulation Command

.TRAN 0.1n 100n

* Probes

.PRINT TRAN V(out1) V(out2)

*.PROBE

.END
```

Layout Simulation Results:

The simulation was performed in LTspice using the exported SPICE netlist from Electric.

Test Cases:

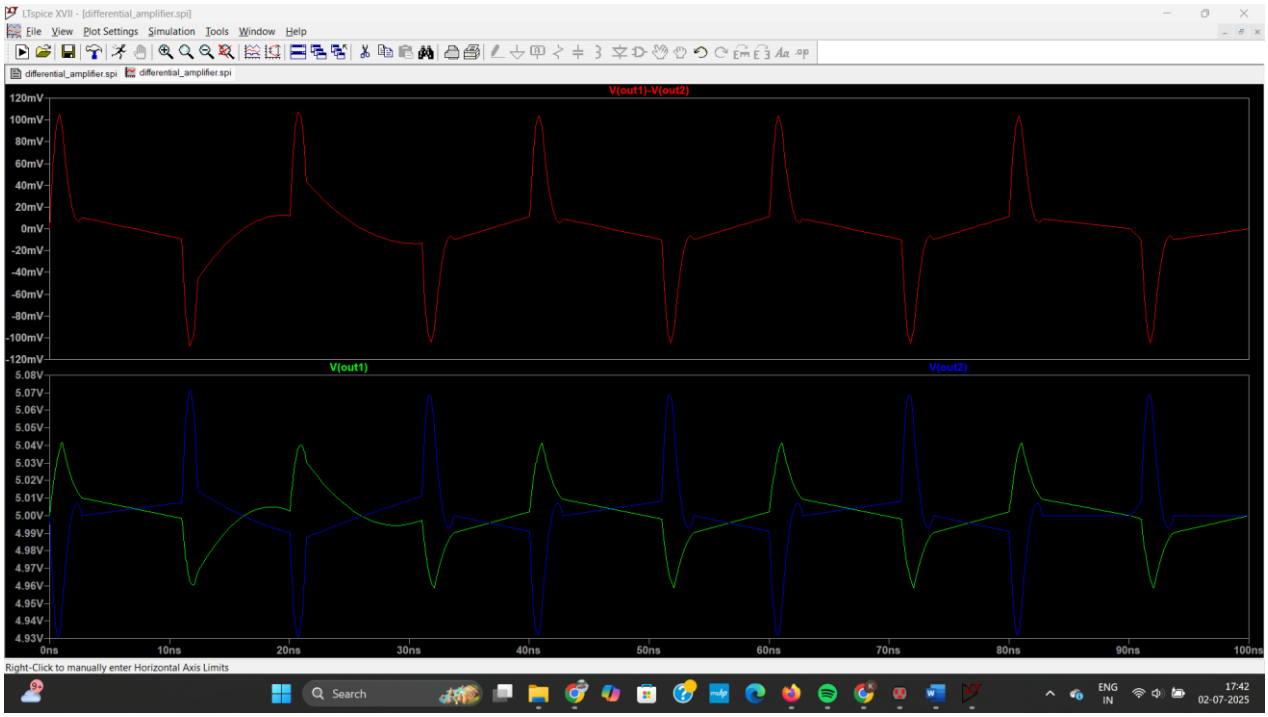
- Differential mode: VIN1 = PULSE(0 0.2V), VIN2 = PULSE(0.2 0V)
- Common-mode test: VIN1 = VIN2 = PULSE(0 0.2V)

Results:

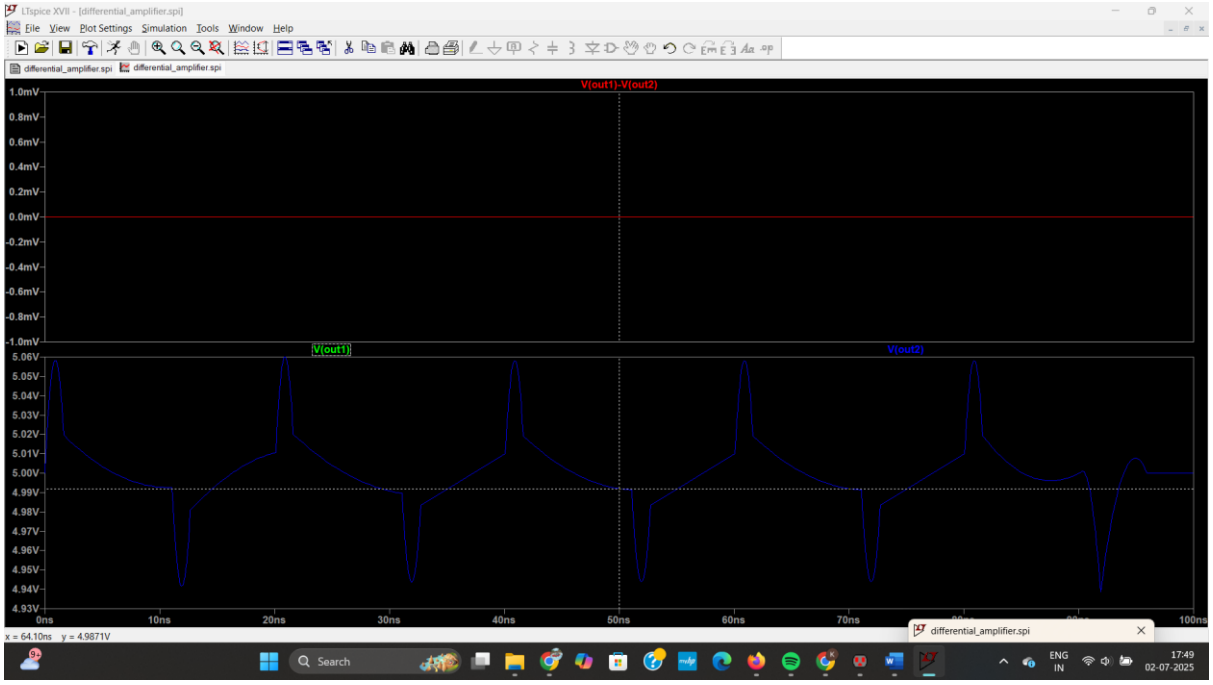
- In differential mode, a clear voltage difference is observed between Vout1 and Vout2.
- In common-mode, both outputs remain nearly constant indicating good common-mode rejection.

Waveforms: Time vs. V(out1), V(out2), and differential voltage (Vout1 - Vout2)

Differential Mode:



Common Mode:


LTspice XVII - [differential_amplifier.spi]
File View Plot Settings Simulation Tools Window Help
differential_amplifier.spi
differential_amplifier.spi
V(out1)-V(out2)
1.0mV
0.8mV
0.6mV
0.4mV
0.2mV
0.0mV
-0.2mV
-0.4mV
-0.6mV
-0.8mV
-1.0mV
V(out1)
V(out2)
5.06V
5.05V
5.04V
5.03V
5.02V
5.01V
5.00V
4.99V
4.98V
4.97V
4.96V
4.95V
4.94V
4.93V
0ns
10ns
20ns
30ns
40ns
50ns
60ns
70ns
100ns
x = 64.10ns y = 4.9871V
differential_amplifier.spi
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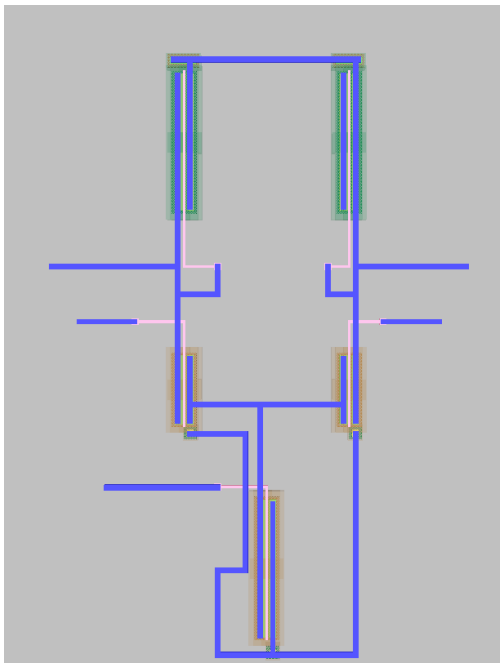
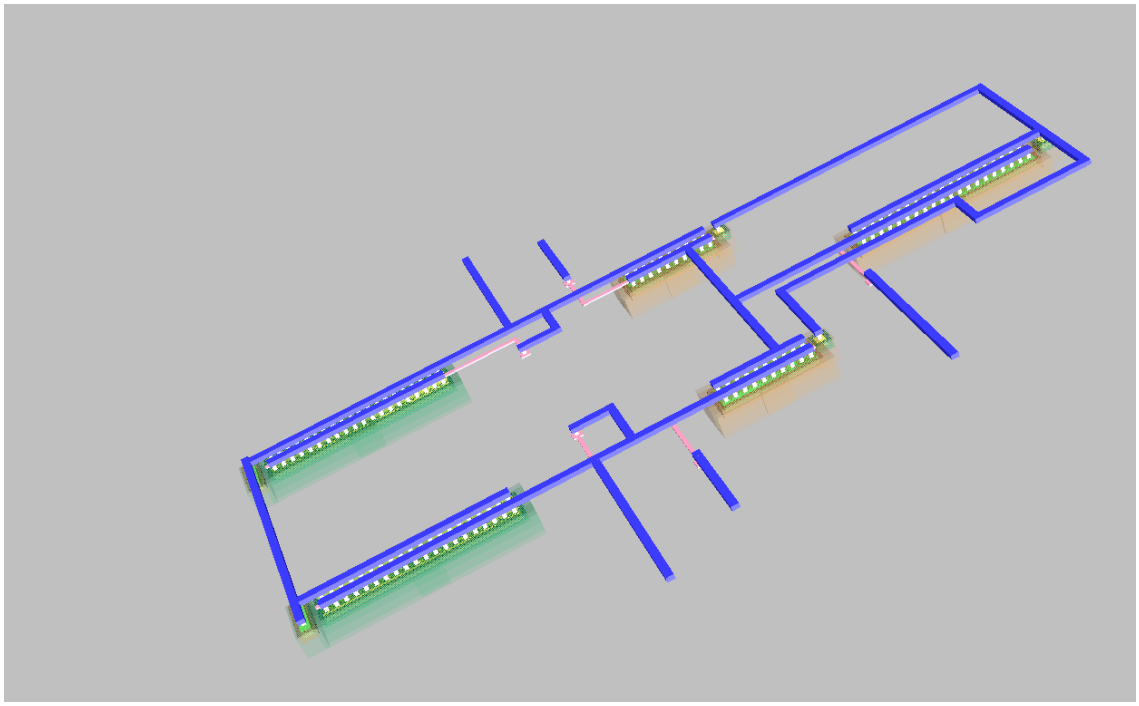
3D Model

A 3D representation of the layout was visualized:

- Layers like metal1, poly, diffusion visualized in stacked format
- Helped understand physical implementation and possible crosstalk/isolation
- Useful for presenting CMOS layers during reviews





Project Outcome

- Gained a comprehensive understanding of differential amplifier operation through schematic and layout design
- Developed skills in analog circuit simulation and analysis using LTspice
- Learned to extract and verify post-layout netlists using Electric EDA tool
- Demonstrated proficiency in handling CMOS design flow from schematic to 3D visualization
- Built a strong foundation for future analog and mixed-signal VLSI design projects